

RYAN CHEN RUI YANG

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EDUCATION

National University of Singapore

Aug 2023 - Expected May 2027

Bachelor of Computing (Hons.) Computer Science, GPA 4.45/5.00

- Modules taken: Operating Systems, Software Engineering, Data Structures and Algorithms, AI & Machine Learning
- **Distinction** in Networks and Distributed Systems Focus Area
- AY24/25 NUS Raffles Hall Videography Club, Vice Chairperson

WORK EXPERIENCE

DBS, Group Infrastructure & Cloud, Singapore (Incoming)

May 2026 - Oct 2026

- Developing end-to-end monitoring and reporting dashboards, and building automated data workflows using Python, Angular, MySQL, and Node.js to improve operational visibility and streamline enterprise processes.

NUS, SWE Undergraduate Teaching Assistant, Singapore

Aug 2025 - Dec 2025

- Mentored students in Java, explaining Git, CI/CD pipelines, SDLC, software testing, and software architecture for scalable and efficient systems, achieving a 100% pass rate with B+ or better scores, and a 4.5/5.0 tutor rating.
- Contributed to refinement and development of course materials, including walkthrough guides and sample code.
- Reviewed and graded student code against established coding principles, providing timely feedback to reduce errors in future submissions by about 70%.

HTX, Cloud & Software Engineer Intern, Singapore

May 2025 - Aug 2025

- Implemented a REST API framework in Microsoft Power Automate to automate image ingestion pipeline for HTX HR, reducing manual data entry by 80% every week, and ensuring 100% availability for continued usage.
- Automated tracking of external contractor leave, connecting a SharePoint data source with an external interface, eliminating 20 hours spent monthly on data syncing and email tracking across departments.
- Created Agentic RAG agents specialised for custom workflows to improve manual analysis of documents, cutting repetitive processing tasks by 50%.

PROJECTS

Automated DevOps Incident Responder

Dec 2025 - Jan 2026

- Architected an autonomous SRE agent using LangGraph, implementing a workflow that iteratively diagnoses runtime errors, queries documentation (Tavily API), and synthesises fix proposals.
- Engineered robust data pipelines by enforcing strict JSON schemas and fallback parsing strategies, eliminating LLM format hallucinations and ensuring system reliability.
- Developed Human-in-the-Loop checks and file system allow-listing (blocking .env/credential access) to prevent unauthorised command injection.
- Implemented a comprehensive Test-Driven Development (TDD) strategy using Pytest and Mocks, isolating LLM logic for zero-cost CI/CD verification of routing and safety protocols.

EcoTrack, Team Co-lead, NUS LifeHack 2025 Hackathon Winner

Jun 2025 - Jul 2025

- Built an AI-driven "Wikipedia of product lifecycles" integrating a Chrome extension, interactive dashboard, and Telegram bot to generate community-verified sustainability scores and lifecycle summaries within a 48-hour timeframe.
- Engineered backend pipeline with OpenAI embeddings, Postgres (vector storage), and a custom tree-based algorithm to scale lifecycle summarisation, maintain performance with sub-500ms latency, and handle 1000+ events per product.

TsFraudPmo, Team Co-lead, DSTA BrainHack Hackathon Finalist

May 2025 - Jun 2025

- Full-stack development of a real-time scam detection platform, integrating a React Native mobile app and Telegram bot to enable instant user reporting and rapid dissemination of verified scam intelligence.
- Implemented custom LLM pipelines, utilising prompt engineering, semantic embedding, and real-time input processing, transforming raw user reports into over 100 validated insights and concise summaries.
- Collaborated closely as a 3-person team alongside a mentor to deliver a production-ready system under competition constraints, demonstrating strong product focus and cross-functional coordination.

TECHNICAL SKILLS

Languages: Python, C++, C#, C, Java, SQL, TypeScript, HTML, CSS

Frameworks & Libraries: React, LangChain, LangGraph, FastAPI

Developer Tools: GitHub, Gradle, CodeCov, Supabase, Tavily

Cloud & Platforms: AWS, Azure, Microsoft Power Platform, Google Cloud Vertex AI